

COMP3242 Week 1 Practice Exam (Questions)

Instructions

Answer all questions. Show key steps for computational questions.

Q1: Tensor Shapes (Concept + Calculation)

A mini-batch of RGB images has shape $(B, C, H, W) = (16, 3, 32, 32)$.

- (a) What is the shape after flattening each image to a vector?
- (b) A linear layer maps each flattened vector to 100 outputs. What is the weight matrix shape and bias shape?
- (c) What is the output batch shape?

Q2: Matrix-Vector Multiplication

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Compute $y = Ax + b$.

Q3: Inner Product and Norm

Let $u = [1, -2, 2]^T$ and $v = [3, 0, -1]^T$.

- (a) Compute $u^T v$.
- (b) Compute $\|u\|_2$.

Q4: Broadcasting

Given tensors $X \in \mathbb{R}^{8 \times 4}$ and $b \in \mathbb{R}^4$, explain why $X + b$ is valid in PyTorch and state the output shape.

Q5: PyTorch Debugging

Consider the code:

```
logits = model(x)                # shape (32, 10)
labels = labels.float()          # labels originally class indices
loss = torch.nn.CrossEntropyLoss()(logits, labels)
```

Identify the bug and provide a correct fix.